

# MICROPULSE

Performance Intelligence Platform for Elite Football

## Platform Technical Brief

A page-by-page walkthrough for medical and performance staff

*Head of Medical · Head of Performance · Sport-Science Lead*

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## Contents

|                                                                      |    |
|----------------------------------------------------------------------|----|
| 1. Executive Summary                                                 | 3  |
| 2. Sport-Science Basis                                               | 4  |
| 3. Architecture Overview                                             | 6  |
| 4. Today Tab — Morning Command Centre                                | 9  |
| 5. Squad Tab — Per-Player Drilldown                                  | 13 |
| 6. Load Tab — sRPE, ACWR & Per-Player Lookup                         | 15 |
| 7. GPS Tab — External Load Intelligence                              | 18 |
| 8. Match Day (MD) Tab — Game-Week Periodisation                      | 19 |
| 9. Intelligence Pages — Indoor Load, Decel, Quadrant                 | 20 |
| 10. Specialist Tabs — VALD, Strength, Volatility, RTP, Trend, Drills | 22 |
| 11. Injuries Page & Decision Engine Deep-Dive                        | 24 |
| 12. Player View — Athlete Self-Service                               | 27 |
| 13. Calibration & Validation                                         | 28 |
| 14. References                                                       | 29 |

# 1. Executive Summary

MicroPulse is a daily decision-support platform for elite football. It closes the four-step monitoring cycle defined by Gabbett (2020) — Collect, Analyse, Communicate, Decide — in a single morning view that the staff opens before the first session. Every player gets one verdict (GREEN / YELLOW / RED) with a transparent explanation, an actionable load recommendation, and increasingly with this April 2026 release a forecast for tomorrow.

The platform integrates the four signal families a top-tier club already collects: subjective wellness (player check-in), external load (Catapult or compatible GPS / IMU), neuromuscular load (VALD ForceDecks, GymAware VBT), and autonomic recovery (WHOOP, Polar Team Pro). Each signal feeds a single decision engine — there is no "averaging away" of a real concern, and any one of the intelligence layers can escalate a player.

This document walks through every coach-facing page in the platform, the sport-science framework underneath each one, and the new explainability layer (counterfactuals, sequence-aware escalation, calibration, forecast) that we believe is the most important capability differentiator for clubs evaluating decision-support tools in 2026.

## Who this brief is written for

- Head of Medical — sections relevant to injury prediction, return-to-play protocols, baseline drift detection, and verdict transparency are flagged FOR MEDICAL throughout.
- Head of Performance — sections relevant to load periodisation, individualised programming, coach decision support, and ROI proof are flagged FOR PERFORMANCE.
- Sport-Science Lead — sections covering the decision engine, calibration loop, and validation methodology are flagged FOR SPORT SCIENCE.

## How to read this document

Sections 1 to 3 give the framework: what the platform is, the peer-reviewed sport-science basis it is built on, and the data architecture. Sections 4 to 12 walk through every coach-facing tab and page. Section 13 covers the validation and calibration methodology. The reference list in Section 14 cites the underlying literature for every framework used.

Each tab section follows the same structure: what the page shows, the decisions it supports, and the sport-science citations behind it. Tables are used wherever the same information would be a long bullet list. "FOR MEDICAL" / "FOR PERFORMANCE" / "FOR SPORT SCIENCE" tags are scattered throughout to help senior staff jump to the parts most relevant to their role.

## 2. Sport-Science Basis

MicroPulse is built on peer-reviewed frameworks from football and team-sport science. We do not invent our own thresholds where established literature already provides them. Where we extend the literature — for example, the personal-baseline-first verdict logic introduced in April 2026 — we cite the evidence and version-tag the change so the staff can audit which version of the algorithm was active on any given day.

### 2.1 Frameworks used by the verdict engine

| Framework                           | Source                                                               | Where it is used in MicroPulse                                                 |
|-------------------------------------|----------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Personal baselines (z-scores)       | Robertson 2017 (Front. Physiol.); Saw, Main & Gustin 2016 (BJSM)     | Wellness sub-score flagging, hybrid readiness coloring (vs global thresholds)  |
| Acute:Chronic Workload Ratio (ACWR) | Gabbett 2017 (BJSM) — "Training-Injury Prevention Paradox"           | Internal load tab, Indoor ACWR, Quadrant view, training-modifier engine        |
| Deceleration burden                 | McBurnie et al. 2022 (J Sports Sci); Harper et al. 2019 (Sports Med) | Decel Intelligence module (4-dim framework), Indoor McBurnie ratio             |
| Internal:External decoupling        | Akubat et al. 2014 (J Sports Sci); Halson 2014 (Sports Med)          | Composite Load Concern, RPE discrepancy detection                              |
| HID% fatigue trend                  | Harper et al. 2019; Russell et al. 2016                              | Mechanical fatigue detection in Composite Load engine                          |
| Monitoring cycle                    | Gabbett 2020 (Sports Med) — "How much? How fast? How often?"         | End-to-end product structure (Collect → Analyse → Communicate → Decide)        |
| Wellness questionnaire              | McLean et al. 2010 (IJSPP) 5-item Likert; Hooper & Mackinnon 1995    | Player morning check-in (sleep, soreness, stress, mood, fatigue)               |
| Neural load (CMJ / NM fatigue)      | Claudino et al. 2017 (J Sci Med Sport); Watkins et al. 2017          | Neural Volatility Intelligence, VALD ForceDecks tab, CMJ recommendation engine |

### 2.2 What we add on top of the literature

Three layers in the platform are MicroPulse extensions of the published frameworks. They are flagged in code with the version of the published framework they extend, and validated by the calibration loop described in Section 11.

- **Personal-baseline-first hybrid coloring** — combines the global Robertson thresholds with each player's own 28-day mean and SD. The verdict only escalates when both the global and personal lenses agree, and only de-escalates when the personal-z is clearly normal for that athlete. This avoids the well-documented false-yellow trap (Saw 2016) where a player whose personal norm sits inside the global YELLOW band can never reach GREEN.
- **Sequence-aware escalation** — stateless single-day inference cannot see chronic patterns. We persist every verdict to athlete\_decision\_history, then escalate the current day to RED when a

player has been YELLOW three days running, and tag SUSTAINED\_RED when red persists two days. This catches the *she-has-been-yellow-all-week* pattern that any sport scientist would catch by eye but that day-by-day automation misses.

- **Counterfactual explanations** — for every non-green verdict, the engine computes up to three single-lever flips that would have improved today's outcome (e.g. *If indoor composite had been TYPICAL → YELLOW not RED*). Honesty constraints are baked in: only single-lever flips, only strictly-improving flips, and when no single change would help (a true multi-concern day) the engine returns empty.

### 3. Architecture Overview

MicroPulse is a single web application backed by a Postgres database (Supabase). It runs as a Progressive Web App on coach laptops, tablets, and phones. There is no native client to install. Player check-in is a 30-second mobile flow that takes five Likert ratings on the morning of training.

#### 3.1 Daily decision cycle

Every page in MicroPulse maps to one step of the Gabbett 2020 cycle:

| Step        | What happens                                                                                                                    | Coach-facing surface                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Collect     | Player wellness check-in (5-item Likert), GPS/IMU upload from Catapult or Polar, CMJ from VALD, sleep/HRV from WHOOP            | Player check-in (mobile), Integrations dashboard, automated overnight sync                     |
| Analyse     | Personal-baseline z-scores, ACWR, decel burden, neural composite, indoor composite, composite load concern                      | Internal Load, GPS, Indoor Load, Decel Intelligence, Quadrant, Volatility, VALD, Strength tabs |
| Decide      | Single-engine verdict (GREEN / YELLOW / RED), session-mode (full / modified / recovery), load action, sequence-aware escalation | Today tab, Squad tab, per-player accordion drilldown                                           |
| Communicate | Daily Briefing card, plain-language player view, coach-to-player chat, broadcast messaging                                      | Daily Briefing on coach Today tab, Player view, Messages tab                                   |

#### 3.2 Data sources and integration points

MicroPulse is signal-agnostic. The platform is currently in production with the following live integrations and can be extended to additional providers without changes to the decision engine — adding a provider is a matter of writing one normaliser:

| Provider        | Signals consumed                                                                                          | Integration mode                          |
|-----------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Catapult Vector | Total distance, HSR, sprint count (V6), accel/decel by band, player load, IMU free-running indoor metrics | REST API, automated daily sync            |
| VALD ForceDecks | CMJ jump height, mRSI, asymmetry %, propulsive impulse, NM fatigue composite                              | VALD Hub API, on-demand and daily sync    |
| WHOOP           | Sleep score, recovery score, HRV (rMSSD), strain, day-strain band                                         | WHOOP API + per-athlete OAuth, daily sync |
| Polar Team Pro  | HR-derived training load, recovery, HRV (selected players)                                                | Polar API, OAuth, daily sync (April 2026) |
| GymAware VBT    | Mean velocity, peak velocity, VBT readiness vs personal exercise baseline                                 | CSV import per session                    |
| WIMU PRO        | GPS+IMU like Catapult, indoor-capable, FMP free-running metrics                                           | CSV import (UI in pilot)                  |
| Hudl video      | Match XML for context-tagging of GPS spikes                                                               | XML upload                                |
| Manual entry    | RPE, soreness areas, illness, injury events,                                                              | Coach UI + player UI                      |

| Provider | Signals consumed      | Integration mode |
|----------|-----------------------|------------------|
|          | return-to-play stages |                  |

### 3.3 Privacy and data ownership

All raw signals stay in the club's Supabase project. MicroPulse never sells, anonymises for resale, or leaves the EU region. Data export is a one-click action; delete is GDPR-compliant cascading delete. Players have a separate privacy view where they see exactly which signals are visible to which staff member. This matters in EU jurisdictions with active works-council oversight of athlete monitoring.

## 4. Today Tab — Morning Command Centre

The Today tab is the first screen the staff opens at 07:00. It is engineered to answer one question in fifteen seconds: who needs my attention this morning, and why? Everything else can be drilled into from there.

*FOR PERFORMANCE: this is the one screen that determines whether the planned session goes ahead, gets modified, or gets a player swap. FOR MEDICAL: this is where a player who has been quietly drifting toward injury risk is surfaced before they show up at the treatment table.*

### 4.1 Daily Briefing card

The Daily Briefing is a single auto-generated card at the top of the page. It collapses everything the morning team meeting used to cover into a five-minute read. It is computed server-side from the same signals the rest of the dashboard uses, so there is no lag or recomputation.

- **Headline** — "Watch 4 players today" with a context badge if yesterday was an OFF day (avoids misleading day-to-day comparisons after rest).
- **Top attention list** — every YELLOW or RED player with a one-line reason, a numeric score chip (Skor / Comp / PL), driver chips that explain the cause (e.g. svefn 2/5,  $\Delta z$  -0.6, comp 0.78), and as of April 2026 a single-line counterfactual hint (e.g. → GREEN If sleep had been GREEN → GREEN).
- **Compliance strip** — how many check-ins and RPE entries are still outstanding for the day, with one-tap reminders.
- **Team pulse** — average readiness, fatigue mix (mechanical / metabolic / global), trend vs yesterday.

#### Why driver chips matter clinically

Each driver chip is computed from the player's personal baseline (28-day window,  $n \geq 7$  observations). Hovering shows the full personal-norm context: "His norm 4.1/5 (SD 0.4) — today 1.8 SD from norm." When the player has insufficient personal history, the system falls back to the published Likert cutoff ( $\leq 2/5$ ) and labels the chip explicitly. Chronic-low chips carry a separate slate badge so a player whose personal norm itself sits at 2.5 doesn't get treated like an acute concern — that's a long-term watch, not a today decision.

### 4.2 Verdict Accuracy card (calibration loop)

Below the briefing sits a small accuracy widget — three tiles for GREEN / YELLOW / RED, each with the percentage of verdicts in the last 30 days that were validated against the next day's outcome and 7-day forward injury data. This is the calibration loop described in detail in Section 11.

- **GREEN safety** = (no escalation, no injury within 7 days) ÷ scoreable verdicts. Higher is better — proves the system safely cleared players who really were ready to train.
- **YELLOW specificity** = (persisted, escalated, or injured) ÷ scoreable verdicts. Higher = correctly flagged players who needed monitoring.



- RED precision = (persisted, escalated, OR injured within 7d) ÷ scoreable verdicts. The interpretation here is intentionally cautious — a recovered RED is reported separately because the recovery may have been driven by the modification the coach applied.

If the team has fewer than 30 scoreable verdicts in the period, the widget is hidden and replaces itself with "Need more data" — we do not show noisy percentages.

### 4.3 Today Command Centre

Below the briefing is the operational workspace where the staff actually executes the morning. It is intentionally separated from the briefing because the briefing is for reading and the command centre is for acting:

- Per-player flag with one-tap recolouring for coach overrides (audit-trailed).
- Locked-card mechanism — once a verdict is reviewed and finalised the row locks, preventing accidental changes after the session has started.
- Bulk "send modified plan" action that triggers chat messages to all flagged players in one click.

### 4.4 Decision Summary card

Sitting between the Command Centre and the Quadrant view, the Decision Summary is a per-player table that condenses every relevant signal into a single row. It is the core operational artefact of the morning — the staff scans it top-to-bottom, then drills into anything that needs attention.

#### Per-row content

- Player name + final colour chip + recommended session mode (Full / Modified / Recovery).
- Numeric STEN, today's z-score,  $\Delta z$  vs yesterday, and ACWR.
- Coach action — auto-derived prescription (e.g. "Pull from high-intensity small-sided games, retain technical and possession"), editable inline.
- Tap-to-open detailed modal with the full Why / What if / Tomorrow's outlook drilldown described in Section 5.

#### 4.4.1 Readiness ↔ Load context strip (per row)

Every row carries a compact strip directly under the colour chip that exposes the four readiness signals (sleep, soreness, stress / mood, energy / fatigue) alongside the four load signals (today's PL spike, ACWR, Catapult intensity band, sRPE if present). This is where decoupling is visible: a player with high readiness but acute external load is treated very differently from one with low readiness and low external load — both might be GREEN by the headline verdict, but the prescription is opposite.

- **Why this matters clinically** (FOR MEDICAL): the Akubat 2014 internal:external decoupling framework shows that mismatch between subjective readiness and objective external load is a stronger short-term injury predictor than either signal alone. The strip surfaces it without forcing a chart drill-down.

#### 4.4.2 Detail modal — Readiness & Load context

Tapping a row opens a structured modal with two side-by-side panels — Readiness items (sleep z, soreness z, stress, energy, plus the chronic-low badges) on the left, Load items (PL spike, ACWR band, decel exposure, indoor composite if applicable) on the right. Each item carries its personal-norm context ("his norm 4.1 / 5") and a colour chip matching the verdict ladder. A note line at the bottom (e.g. "Sleep z -2.1 with PL spike 1.45 — high decoupling, recommend technical session") translates the data into a sentence the staff can act on.

#### 4.5 Readiness × Load Quadrant (Today tab)

At the foot of the Today tab sits the live Readiness × Load Quadrant — a Gabbett 2017 four-quadrant scatter plot that places every player on two axes simultaneously: x-axis = external load proxy (PL spike or ACWR), y-axis = subjective readiness (STEN). Quadrant labels are deliberately operational, not statistical:

| Quadrant                             | Profile                                                                      | Recommended action                                                     |
|--------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Top-right (HI load, HI readiness)    | Adapted, ready, just trained hard. Highest absolute injury risk on volume    | Maintain, but cap volume; protect the recovery window                  |
| Top-left (LO load, HI readiness)     | Fresh, undertrained — biggest opportunity for stimulus                       | Push: this is the player who can take the hard session                 |
| Bottom-right (HI load, LO readiness) | Accumulated load + low readiness — classic injury danger zone (Gabbett 2017) | Force recovery, modify session, consider RTP-style monitoring          |
| Bottom-left (LO load, LO readiness)  | Detrained / illness / chronic stress — the silent under-loader               | Investigate cause before increasing load (illness, sleep, life stress) |

The Today-tab quadrant is the live, today-only view. The deeper analytics quadrant (Section 9.3) shows the same axes over a 7-day or 28-day window with trend arrows — useful for the weekly periodisation review. Both views read the same underlying signals so the numbers always match.

*FOR PERFORMANCE: the bottom-right quadrant is the single most actionable cell in the platform. If a player lands there two days running and the sequence rule (Section 11.2.2) hasn't yet escalated them to RED, the staff can act ahead of the system.*

## 5. Squad Tab — Per-Player Drilldown

The Squad tab is the operational screen where the staff actually sits down and reviews every flagged player one by one. It is the deepest verdict-explanation surface in the platform. Every per-player drilldown is reachable from the Today tab in one click, but the Squad tab presents the full roster in a scannable list with the same drilldown available inline.

*FOR MEDICAL: this is where you should expect to spend the first 20 minutes of every morning. Every layer of the verdict — readiness, neural, injury risk, indoor load, decel intelligence — is exposed as an expandable section so you can audit any decision the engine made.*

### 5.1 Player row at a glance

- Final colour chip (with override marker if a coach changed it).
- STEN score (0-10 standard ten — Robertson 2017 normalisation of the 28-day baseline).
- $\Delta z$  vs yesterday — surfaces overnight changes even when both days are still inside the same colour band.
- Composite chip (load concern), PL spike chip (player-load  $\times$  28-day average), fatigue type tag (mechanical / metabolic / global).

### 5.2 Show-Details accordion

Clicking Show Details expands six sections, each independently collapsible. We deliberately use accordions instead of tabs because medical and performance staff sit together at the screen and need to look at multiple sections at once during the same conversation.

#### Readiness Decision section (most-used)

- **Why panel** — bulleted list of the dominant reasons today's verdict came out the way it did, in plain language. Streak escalations (e.g. "3rd yellow day in a row — escalating to RED") appear as the top reason when they fired.
- **Coach Action panel** — actionable recommendations (full / modified / recovery) plus a "Why?" button that opens the full Performance Intelligence drawer.
- **Tomorrow's outlook panel** (NEW April 2026) — three scenarios (If improves / If holds / If declines) with a confidence chip (low / medium / high). The "If holds" scenario applies the same streak-escalation rule that produced today's verdict, so a player on day 2 of yellow correctly forecasts to RED for tomorrow with high confidence. Bilingual rationale lines explain which gate is dominant (streak rule firing vs trajectory).
- **What If panel** (NEW April 2026) — up to 3 single-lever counterfactuals with the resulting hypothetical state. Coach immediately sees what would need to change to flip the verdict. When the engine returns empty (no single change would help), that is itself information — it means the player is genuinely multi-concern.

#### Injury Risk section (FOR MEDICAL)

Independent rule-based engine that runs alongside the readiness decision. It uses the same wellness signals plus VALD asymmetry, accel:decel ratio, HSR/sprint-count spikes, and the chronic personal-low flags. Outputs LOW / MODERATE / HIGH plus a separate confidence band, the contributing rules,

and a recommendation list. Critically, this engine never overrides the readiness decision — it raises an independent voice that the verdict engine's gate-logic then folds in as one input. This separation makes it auditable and lets the medical staff disagree with the verdict without disagreeing with the injury-risk reasoning.

### **Readiness Inputs section**

Numerical view of every signal that fed today's verdict — total score, sleep, soreness, energy, stress, mood, ACWR, HSR, max velocity %, neural composite, residual MLI, residual decel, plus personal-baseline z-scores for each one. This is the staff's audit trail when they need to disagree with the engine and write a justified override.

### **Performance Intelligence section**

The full drawer that the Today tab Daily Briefing summarises — risk band, load action, main driver, prescription, neural volatility lines, final recommendation, session-draft preview if a session is being built for the day. Sport scientists use this for case-conference reviews.

### **Final Recommendation section**

Single line in the format the staff would actually write on a whiteboard: "Modified — pull from high-intensity small-sided games, retain technical and possession." Auto-generated, editable inline, audit-trailed.

## 6. Load Tab — sRPE, ACWR & Per-Player Lookup

The Load tab is the staff's deep-dive workspace for subjective and historical load monitoring. It is built around the Acute:Chronic Workload Ratio model (Gabbett 2017) but extends well beyond a single chart — five distinct cards on this tab cover everything from the per-player historical lookup that medical staff use during return-to-play planning to the team-level ACWR risk overview that sport scientists use for weekly periodisation review.

### 6.1 Player Historical Snapshot card

A single search-and-select control exposes any player's longitudinal record. Pick a player and the card renders, in one scrollable view:

- Per-day readiness colour history (rolling 28 days), plus the underlying STEN, z-score, and  $\Delta z$ .
- Personal-baseline overlay — mean and  $\pm 1$  SD for every wellness sub-score, with the player's actual values plotted against their own normal range.
- Sleep, soreness, stress, mood, energy time series — small-multiple sparklines with the chronic-low band shaded if the player's personal mean for that signal sits below the threshold.
- Load history — sRPE per session, plus PL spike vs 28-day baseline.
- Injury and illness markers overlaid on the timeline so the staff can see the antecedent signals to any past event.

*FOR MEDICAL: this is the card that replaces the spreadsheet most clubs still keep on the player's history. Auditable, exportable to PDF, and always backed by the underlying signals — not free-text notes that drift over a season.*

### 6.2 Session RPE monitoring card

Per-session table covering the rolling 28-day window. Each cell is a session  $\times$  player intersection, coloured by sRPE (RPE  $\times$  duration in minutes). Designed for the post-session review the staff runs at the end of training:

- RPE compliance row at the bottom — fraction of players who logged RPE for each session, with cells under 70% highlighted (data-quality guard).
- Per-session team average row, plus individual deviations from that average.
- Discrepancy flag — when an individual's RPE diverges from team average by  $> 1.5$ , the cell carries a small triangle and the player is queued for a brief follow-up. Catches both under-reporting ("I want to play through") and over-reporting ("something is going on off-field").

### 6.3 Daily Internal Load card

Today-only summary view that sits side-by-side with Session RPE. Aggregates today's check-in + RPE into a single per-player line so the staff can confirm at a glance: who logged, what their internal-load reading was, and how it compares with their personal 7-day rolling average. Used immediately after a session to confirm completeness before the staff leaves the building.

## 6.4 ACWR Risk Overview card (Internal Load)

Team-wide ACWR matrix — every player on a single grid with their acute (7-day rolling sum), chronic (28-day rolling average), and ACWR ratio. The card uses the Gabbett 2017 sweet-spot band 0.8 – 1.3 as the safety zone:

| ACWR band          | Interpretation                                                    | Action                                                                |
|--------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------|
| < 0.5              | Severe undertraining — detrained, ill, or returning from a layoff | Plan a structured ramp-up, not a sudden return to full load           |
| 0.5 – 0.8          | Below sweet spot — slight under-load risk                         | Increase volume gradually; tolerate the cell if intentional           |
| 0.8 – 1.3 (green)  | Sweet spot — adapted, lowest injury risk per Gabbett 2017         | Maintain. No action required                                          |
| 1.3 – 1.5 (yellow) | Elevated — early acute-spike pattern                              | Monitor; reduce next session intensity if other signals also flagging |
| > 1.5 (red)        | Acute spike — soft-tissue injury risk doubled (Gabbett 2017)      | Force recovery; modify or pull from the next high-intensity session   |

- **Critical caveat** (FOR SPORT SCIENCE): the Bowen 2017 critique correctly notes that ACWR in isolation is a weak injury predictor and the original 2017 effect sizes have not always replicated. MicroPulse uses ACWR as one input to the verdict engine — never as a standalone alarm. The card explicitly labels the band but defers escalation to the multi-signal Decision Engine described in Section 11.

## 6.5 External Load summary (GPS) card

On the Load tab we also surface an external-load summary card that links across to the full GPS tab (Section 7). This is intentional — many staff workflows start on the Load tab and want a quick "what does Catapult say about yesterday" answer without flipping tabs. The card shows:

- Yesterday's team average for total distance, HSR, sprint count V6, accels/decels, and player load.
- A coloured intensity tag (LOW / MEDIUM / HIGH / OFF) so the staff can immediately classify the session.
- Per-player thumbnails with PL spike chips for outliers.
- Drill-in link to the full GPS tab for analyst-level breakdowns.

## 6.6 What the Load tab supports as decisions

Three concrete coach decisions are anchored on this tab:

1. Weekly periodisation review — head coach, head of performance, and sport scientist sit together with the ACWR Risk Overview and decide the next week's volume profile per player.
2. Return-to-play history check — medical staff opens the Player Historical Snapshot to verify the loading pattern that preceded a previous injury before clearing the player.
3. End-of-session debrief — Session RPE + Daily Internal Load cards are reviewed at the end of training; missing or discrepant entries are chased before the staff leaves.



## 7. GPS Tab — External Load Intelligence

The GPS tab is where every Catapult or WIMU PRO daily file lands. It is the most data-rich screen in the platform and the home base for sport scientists analysing periodisation week to week.

### 7.1 Per-player section

- Today vs 28-day baseline for the eight standard external-load metrics (total distance, HSR distance, sprint count V6, accels/decel by band 2-3 / 4-5+, player load, max velocity).
- Per-metric chips coloured by today / baseline ratio — light, typical, elevated, heavy, spike.
- Player-load spike (today / 28-day average) prominently displayed — Gabbett 2010 showed this is the single most predictive external-load metric for next-day fatigue.
- Sprint count chip (V6 efforts) for sprint-mechanics monitoring (Buchheit 2010).

### 7.2 Team aggregate section

- Team average per metric for the day, plus standard deviation.
- Today vs team — outliers above 1.5x team average are flagged so the staff sees who carried the session disproportionately.
- Drag-and-drop column reorder so the staff can put the metrics they care about most in the leftmost columns (preference is per-user, persisted).

### 7.3 Composite Load Concern (FOR PERFORMANCE)

Independent composite engine that combines:

- Neuromuscular Burden Score (NBS) — Catapult-derived composite of HSR, sprint count, and high-band accels (Verheul 2020).
- Mechanical Load Index (residual MLI) — accel:decel coupling residualised against player-load (in-house extension of McBurnie 2022).
- Metabolic Load Score — distance-density × intensity-band time (Coutts 2018).
- HID% fatigue trend — Harper 2019 high-intensity-distance decline pattern.

Outputs concern level (none / low / moderate / high), fatigue type (mechanical / metabolic / global), and an explanation list of which signal escalated. The fatigue type tag flows directly into the Decision Summary so the staff knows whether to modify with metabolic recovery vs CNS recovery vs mechanical de-load — they are different prescriptions.



## 8. Match Day (MD) Tab — Game-Week Periodisation

The MD tab anchors every week to the match day (MD) and labels every other day relative to it (MD-4, MD-3, MD-2, MD-1, MD, MD+1, MD+2). Every recommendation throughout the platform respects the MD context — a high readiness score on MD-1 is treated very differently from the same score on MD-4.

### 8.1 Match-day setup

- One-tap MD assignment per week — the staff sets which date is MD and the system labels every other day automatically.
- Recovery protocol guards — MD+1 sessions automatically default to recovery mode unless the staff explicitly overrides; CMJ test prompts auto-fire on MD+1 and MD-2.
- Travel and competition-density signals — match congestion and travel load are inputs to the injury-risk engine (Lago-Penas 2010).

### 8.2 Cup or international break handling

- Multiple match days within a week supported — the system handles double-match weeks correctly without the staff having to pick one as canonical.
- International call-up tracking — players away on national duty get a separate state (AWAY / RTP) and their absence is visible in the Squad tab so the staff plans around it.

## 9. Intelligence Pages — Indoor Load, Decel, Quadrant

Three deeper analytics pages sit alongside the main dashboard. Each one answers a question that the daily verdict cannot: not because the verdict is wrong, but because some questions are weekly or monthly questions, and the daily view would clutter if it tried to surface them too.

### 9.1 Indoor Load Intelligence

Built for clubs whose winter / off-season training happens indoors. GPS does not work indoors, so external-load monitoring needs an alternate signal pipeline. We use the IMU-derived Free-running Metabolic Power (FMP) framework with seven intensity bands plus an eight-band IMA (Inertial Movement Analysis) free-running metric.

- Indoor Composite Score — IMU-only weighted combination of metabolic-power band time, accel/decel exposure, and IMA stride load. Drops the GPS-dependent metrics (HMLD, decel band 2-3) entirely so the score remains valid in basketball-style indoor sessions.
- Indoor McBurnie ratio — adapted from McBurnie 2022, using indoor-only signals. Coupling between deceleration intensity and free-running intensity. Sweet-spot 1-10; outside that band flags a coupling problem.
- Indoor ACWR — same Gabbett 2017 framework applied to the indoor composite score.
- Composite-band classification — light / below\_average / typical / heavy / spike, each tuned against a 28-day baseline of indoor sessions.

Basketball clubs (we have one in pilot) run with `effectiveIndoorMode = true` permanently. Football clubs toggle it on per session — the system can also auto-detect indoor mode if every player on the team has zero GPS data for the day.

### 9.2 Decel Intelligence (McBurnie 2022 4-Dimension Framework)

This module operationalises the McBurnie et al. 2022 review of deceleration as an injury risk factor. Four orthogonal dimensions are computed per player per session:

- Overload exposure — how often the player decelerates above 4 m/s<sup>2</sup> (high-magnitude band). Cumulative over a 28-day window.
- Underload risk — extended periods of low deceleration exposure that compromise the eccentric tissue-tolerance maintained by regular high-decel work (Hewett 2010 underload-recovery loop).
- Decel:accel coupling — ratio of deceleration count to acceleration count. Both <0.7 and >1.3 indicate a coupling problem (player either braking far more than accelerating, or vice versa).
- Decel:sprint coupling — ratio of deceleration count to V6 sprint efforts. Captures the eccentric-decel-after-sprint pattern responsible for hamstring strains (Edouard 2016).

Each dimension outputs green / yellow / red. The overall flag is the worst of the four. The verdict engine treats overall = red as a red gate, overall = yellow as a yellow gate.

*FOR MEDICAL: this is the most evidence-based hamstring-injury early-warning module available outside the published research. The McBurnie group is the same lab whose work is now incorporated into the Aspetar return-to-play guidelines.*

### 9.3 Quadrant View (Gabbett 2017)

Two-dimensional scatter plot — external load (player load or distance) on the x-axis, internal load (sRPE or HRV-load) on the y-axis. Each player is a dot. The high-external-high-internal quadrant (top-right) is the high-injury-risk quadrant per Gabbett 2017.

- Filter by week, by phase, by position group.
- Trend arrows — where each player has moved over the last 7 days.
- Hover for full per-day breakdown.

Quadrant view is deliberately a standalone analytics view — it does NOT feed the daily verdict because the signals it visualises (player-load, sRPE) already feed the verdict via Composite Load Concern and Readiness Decision. Adding Quadrant as a verdict input would double-count.

## 10. Specialist Tabs — VALD, Strength, Volatility, RTP, Trend, Drills

### 10.1 VALD ForceDecks tab

Direct integration with VALD Hub. Imports CMJ, isometric mid-thigh pull, and squat profile data.

Computes:

- Daily NM fatigue composite — jump height + mRSI + impulse + asymmetry, each z-scored against the player's 28-day baseline.
- Asymmetry % — between-leg asymmetry on bilateral jumps; >10% threshold per Bishop 2018.
- CMJ recommendation engine — auto-prompts a CMJ test on (a) MD-2, MD+1 protocol days, (b) when neuromuscular flag goes yellow/red reactively, (c) when CMJ is stale (>7 days since last test).
- Hamstring asymmetry warnings — VALD Nordic-bord output flagged through the injury-risk engine.

### 10.2 Strength tab — Velocity-Based Training (VBT)

Reads GymAware CSV exports per session. Computes mean velocity for each exercise, compares to the player's exercise-specific baseline, and outputs a VBT readiness score. When VBT mean velocity is below personal baseline by more than 10%, the player is flagged as neurally fatigued for the day — an additional gate the verdict engine respects.

### 10.3 Volatility — Neural Volatility Intelligence

Looks at variability in the readiness signal itself, not the level. The intuition (Schiphof-Godart 2018): a player whose readiness oscillates wildly day to day is in worse adaptive shape than one with a low but stable score. We compute coefficient-of-variation in the rolling 14-day window and surface the top 5 most-volatile players. Volatility >0.4 with average readiness <14/25 fires a CAUTION neural status, which is a verdict gate.

### 10.4 RTP — Return to Play

Per-injured-player view that walks through a stage-based protocol: Acute → Sub-acute → Sport-specific reintegration → Return to play → Return to performance. Each stage has criteria (e.g. asymmetry < 10%, jump height within 5% of pre-injury baseline). The view shows where the player is, the next criterion they need to meet, and the projected return date.

- Auto-promote — when a player's verdict is RED for more than 3 days the system asks the staff if this should be coded as illness or injury. Once coded, the player's verdict logic switches to RTP mode and the daily readiness display becomes a stage progression instead of a colour.
- Auto-demote — when RTP criteria are met for 7 consecutive days, the system suggests promoting the player back to active monitoring.

*FOR MEDICAL: the RTP tab and the injury log (Section 11) together replace the spreadsheet that most clubs still use to track injured-player progression. Auditable, timestamped, with the underlying signal data attached at every stage transition.*

## 10.5 Trend tab

Per-player 14-day sparkline view — readiness, sleep, soreness, neural composite, and player-load on a single page. Designed for the weekly periodisation review where the staff sits down with the head coach and walks through the squad. Click a sparkline to drill into the underlying signal history.

## 10.6 Drills tab

Exercise library tied into the session planning workflow. Each drill has tags (intensity band, body region, decel-load profile), an optional SVG diagram (auto-generated from text descriptions), and a usage history per session. Coach can build a session by dragging drills into a timeline; the system computes the projected internal and external load and warns if it would push a flagged player out of their recommended band.

## 11. Injuries Page & Decision Engine Deep-Dive

### 11.1 Injury log + correlation engine (FOR MEDICAL)

Single canonical injury database for the club. Every entry captures injury date, type, body side, mechanism, severity, days lost, return date, and an attached set of retrospective signals — what the verdict engine was saying for that player in the 14 days leading up to the event.

- Injury log entry — one form, takes 30 seconds, no free-text dependency for the analytics.
- Retrospective signal snapshot — automatic at the moment of logging. Captures readiness STEN, z-score, ACWR, decel composite, neural composite, indoor composite for the 14 prior days. This is the proof-of-warning evidence the staff can audit later.
- Correlation analysis — for each injury, the system computes which signals were elevated above 1.5 SD in the 14 days prior. Surfaces patterns: "3 of last 5 hamstring strains were preceded by decel:sprint coupling > 1.5 in the 7 days prior."
- Read-only analytics page — separate from the operational injury log so analysts can pull historical patterns without seeing live coaching information.

### 11.2 The Decision Engine

Every per-player verdict in MicroPulse goes through a single decision builder. This subsection details how it works because verdict transparency is the most-asked-about question from clubs evaluating the platform.

#### 11.2.1 The severity ladder

| Signal source                                               | RED gate                   | YELLOW gate                 |
|-------------------------------------------------------------|----------------------------|-----------------------------|
| Hard block / Rehab / RTP                                    | Any active block           | —                           |
| Readiness Decision (STEN, z-score, $\Delta z$ , volatility) | RED state                  | YELLOW state                |
| Neural Load (CMJ + autonomic + composite)                   | suppressed (CRITICAL/HIGH) | caution (RISING)            |
| Injury Risk (VALD + GPS spike + soreness + chronic-low)     | HIGH                       | MODERATE                    |
| Composite Load Concern (Outdoor: RPE+NBS+Metabolic)         | high                       | moderate                    |
| Indoor Composite Score (FMP / IMU höll-mode)                | spike                      | heavy                       |
| Indoor McBurnie ratio                                       | red (<0.5 or >15)          | yellow (0.5–1 or 10–15)     |
| Indoor ACWR (Gabbett 2017)                                  | red (<0.5 or >1.5)         | yellow (0.5–0.8 or 1.3–1.5) |
| Decel Intelligence (McBurnie 2022 4-dim overall)            | red                        | yellow                      |

Logic: any RED gate fires → verdict = RED. Otherwise any YELLOW gate fires → verdict = YELLOW.

Otherwise the readiness state passes through unchanged. No averaging, no weighting that could dilute a single high-risk signal.

### 11.2.2 Sequence-aware escalation

After the per-day verdict is computed, the engine checks the player's last 7 days of decision history. Two escalation rules apply:

- If the player has been YELLOW for 3 days running (today included), escalate to RED with reason CHRONIC\_YELLOW\_3D.
- If the player is in their 2nd consecutive RED day, keep RED but tag SUSTAINED\_RED\_2D — coach guidance shifts to "rest is non-negotiable".

### 11.2.3 Counterfactual explanations

After the verdict is finalised, the engine re-runs the same decision tree with each input lever flipped to its best value. If the result is strictly better than today's verdict, that lever is reported as a counterfactual. Up to 3 are returned, sorted by impact (RED→GREEN > RED→YELLOW > YELLOW→GREEN). Honesty constraints:

- Single-lever flips only — multi-lever is unactionable.
- Strictly improving only — no "if this had been worse" alarmism.
- Hard block and rehab are excluded levers — they are clinical states, not measured signals.
- When no single lever helps (multi-concern day), the engine returns empty — and that is itself information.

### 11.2.4 Tomorrow's outlook (forecast)

New in April 2026. Three scenarios — If improves / If holds / If declines — projected through the same streak escalation gate that produced today's verdict. The If holds scenario is the most useful: a player on day 2 of yellow correctly forecasts to RED for tomorrow with high confidence because the streak rule will fire. The forecast adds a confidence band (low / medium / high) based on data depth and trajectory consistency. We deliberately do not output a probability — that requires labelled outcome data we are still collecting.

## 12. Player View — Athlete Self-Service

Every staff-facing decision in MicroPulse has a player-facing counterpart. Athletes do not see only their colour — they see the same explanation the staff sees, in plain language, with the option to message their coach directly.

### 12.1 Morning check-in (mobile)

Five-item Likert questionnaire on the player's phone: sleep quality, soreness, fatigue/energy, stress, mood. 30 seconds, designed by McLean (2010) and validated specifically for elite team sport.

Response submission triggers the verdict engine within 200ms; the player sees their colour and a one-line plain-language verdict immediately.

### 12.2 "Why am I not green?" panel

- Plain-language explanation of which factor pulled the verdict down — built from the same explanation engine the coach sees but stripped of jargon.
- Severity tag per factor (high / moderate / low) so the player knows which input mattered most.
- "What would help" panel (NEW April 2026) — same counterfactual engine output, surfaced as a player-actionable list. "If sleep quality had been GREEN → GREEN" tells the athlete which lever they personally control.

### 12.3 Trend & history view

Personal 14-day sparkline of every signal the player has contributed (readiness, sleep, soreness, optionally CMJ if VALD-equipped, optionally HRV if WHOOP-equipped). Players can opt to share specific signals with their coach via the privacy panel — a GDPR-aligned granular consent model that is uncommon at our pricepoint.

### 12.4 Coach chat

Direct messaging to the coaching staff with 7-day retention. Daily verdicts auto-broadcast a contextual message (e.g. "Reduced session today based on your readiness — focus on technical work"). Players can reply, attach photos of injuries, etc.



## 13. Calibration & Validation

A decision-support platform is only as useful as it is calibrated. Three layers of validation are baked into MicroPulse, all visible to the staff and editable by the system administrator. None of them are vanity metrics — they are designed to make the system honest about its own accuracy.

### 13.1 Verdict accuracy widget (live)

The Verdict Accuracy card on the Today tab shows three numbers refreshed daily, computed against the prior 30-day window:

- GREEN safety = (no escalation, no injury within 7d) ÷ scoreable verdicts. The system's calibration when it cleared a player to train fully.
- YELLOW specificity = (persisted, escalated, or injured) ÷ scoreable verdicts. The system's accuracy when it flagged a player for monitoring.
- RED precision = (persisted, escalated, OR injured) ÷ scoreable verdicts, with a separate count of recovered cases reported as a caveat — those may be system-driven recoveries that should be counted as wins.

Sample-size guard: the widget hides the percentages when fewer than 30 scoreable verdicts have been computed ( $n < 30$  makes proportion estimates too noisy). The interpretation footnote in the expanded view explains methodology in plain language.

### 13.2 Calibration view (SQL, exposed to admin)

Two database views are exposed to club analysts who want to slice the calibration data themselves:

- `vw_decision_calibration` — per-day per-player outcome view: predicted state, next-day state, trajectory (persistent / escalated / recovered), injured-within-7d boolean.
- `vw_decision_calibration_summary` — 30-day rolling aggregation per team × predicted state, with breakdown counts (persistent / escalated / recovered / unknown / injury).

These views can be queried directly from any BI tool (Tableau, Power BI, Looker) the club already uses. We export them via the Reporting Centre as a weekly CSV by default.

### 13.3 Outcome-loop (planned for week 8 of pilot)

Every saved verdict is persisted to `athlete_decision_history` with the full input signal snapshot in JSONB. By week 8 of a pilot we have enough data to train a calibration model that maps verdict + input-signals to actual outcomes (persistence, injury, recovery). The first deliverable is a confusion matrix with confidence intervals; the second is a calibration plot showing whether the engine's RED threshold should be tightened or loosened against the club's specific outcome rate.

*FOR MEDICAL: this is the first commitment a vendor at our pricepoint typically refuses to make. We commit to a written calibration report at week 8 and week 12 of any pilot, comparing system verdicts against the club's own injury and availability outcomes. The data and the report are yours to publish if you wish.*

## 14. References

Selected peer-reviewed sources that underpin the platform. The full reference list (40+ papers, including methodology critiques and dissenting evidence) is available on request.

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## Closing

MicroPulse exists to make the day-to-day work of medical and performance staff at football and team-sport clubs measurably better. The platform is in active production use across multiple clubs and continues to evolve based on staff feedback and the calibration loop described in Section 13.

If anything in this brief raises a question — about the methodology, the integrations, the verdict engine, or any specific page — please reach out directly. We respond to technical queries from medical and sport-science staff within one working day.

Thank you for reading.

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